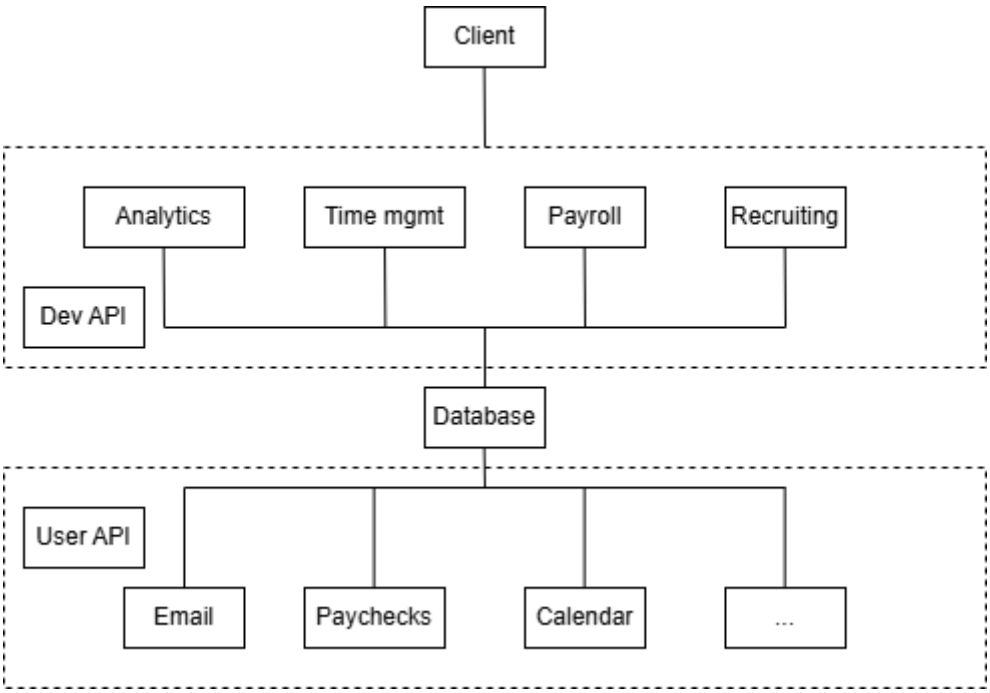
1.

Personio Architecture



2.

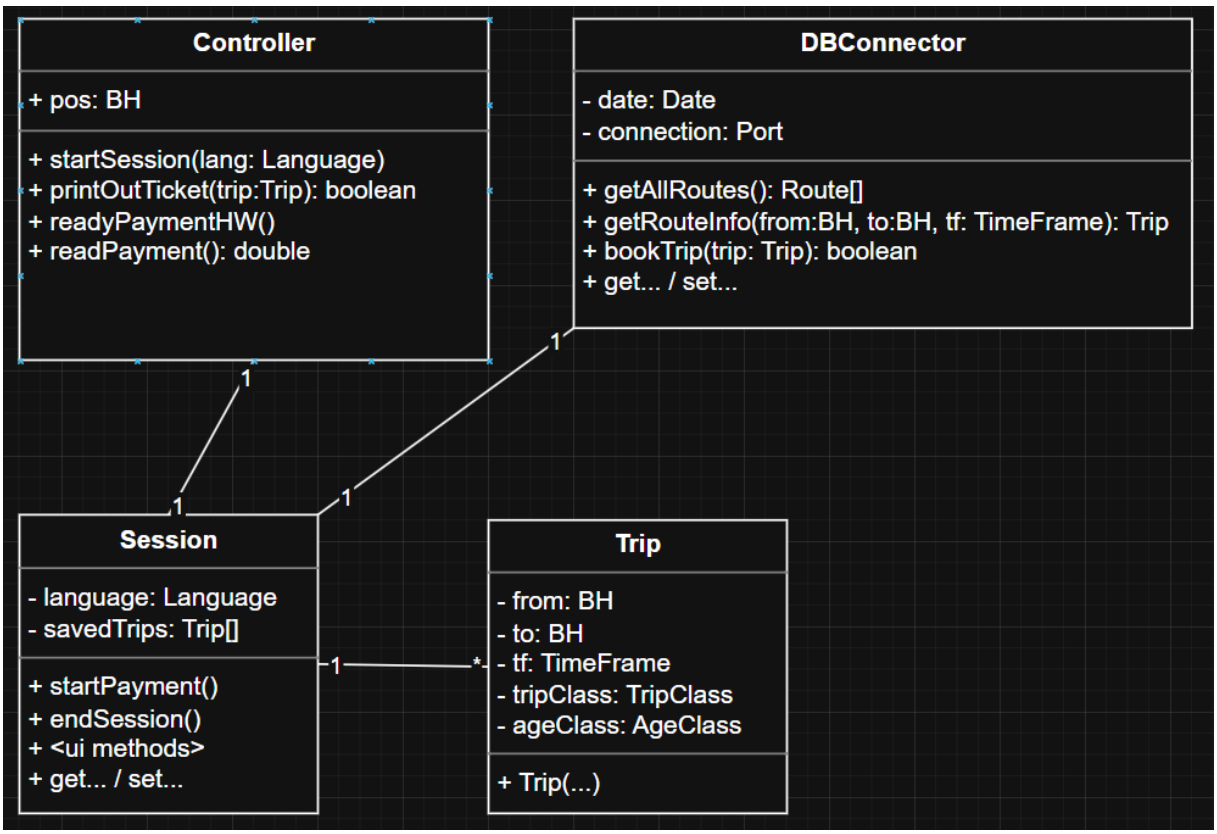
P2P (Peer-to-Peer) Architecture is beeing used, which splits the workload between client and host servers.

Example: Local information like personal preferences, HUD and temporal inventory changes may be stored on the local machine of each user, while important, global data, like player currency amounts, in-game-market information, player position (partly), e.g. are stored on the game providers server.

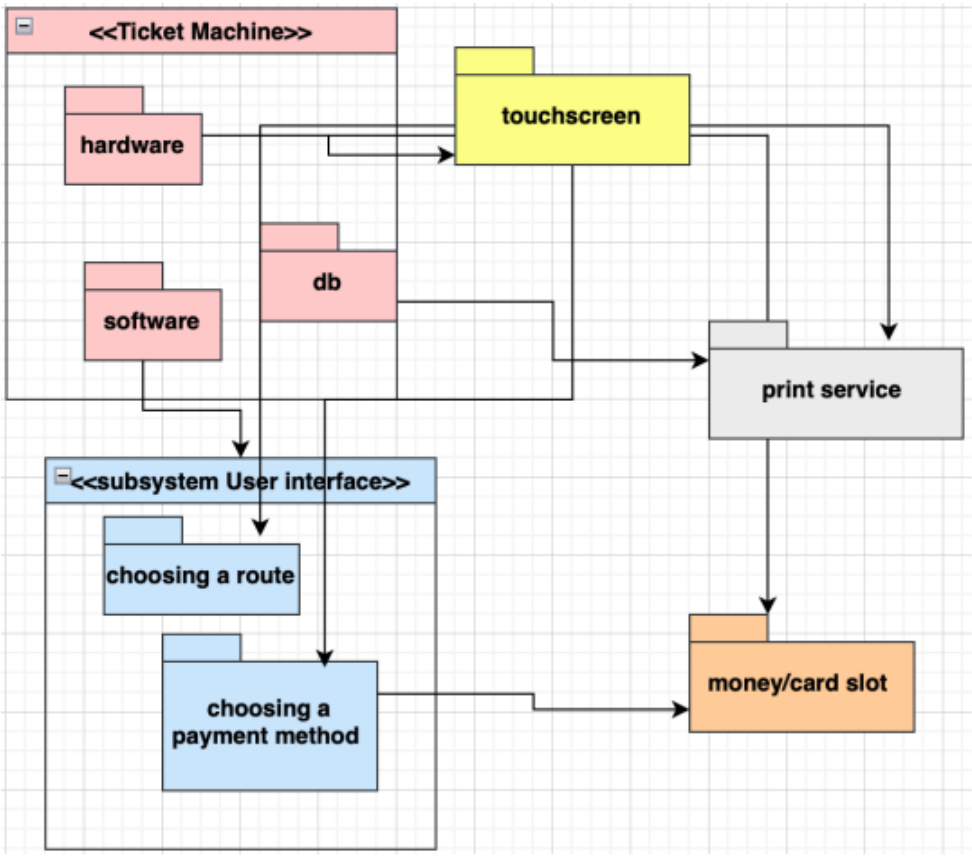
A lot of information is also saved on both, with it beeing synchronized between Server and Client. A good example of this is player position in competetive shooter games. While the Server keeps track of every players movement, the precise position of every player changes a bit for every client, securing a stable enviroment for each player. Altough this makes movement more smooth, this may lead to errors in precision.

3.

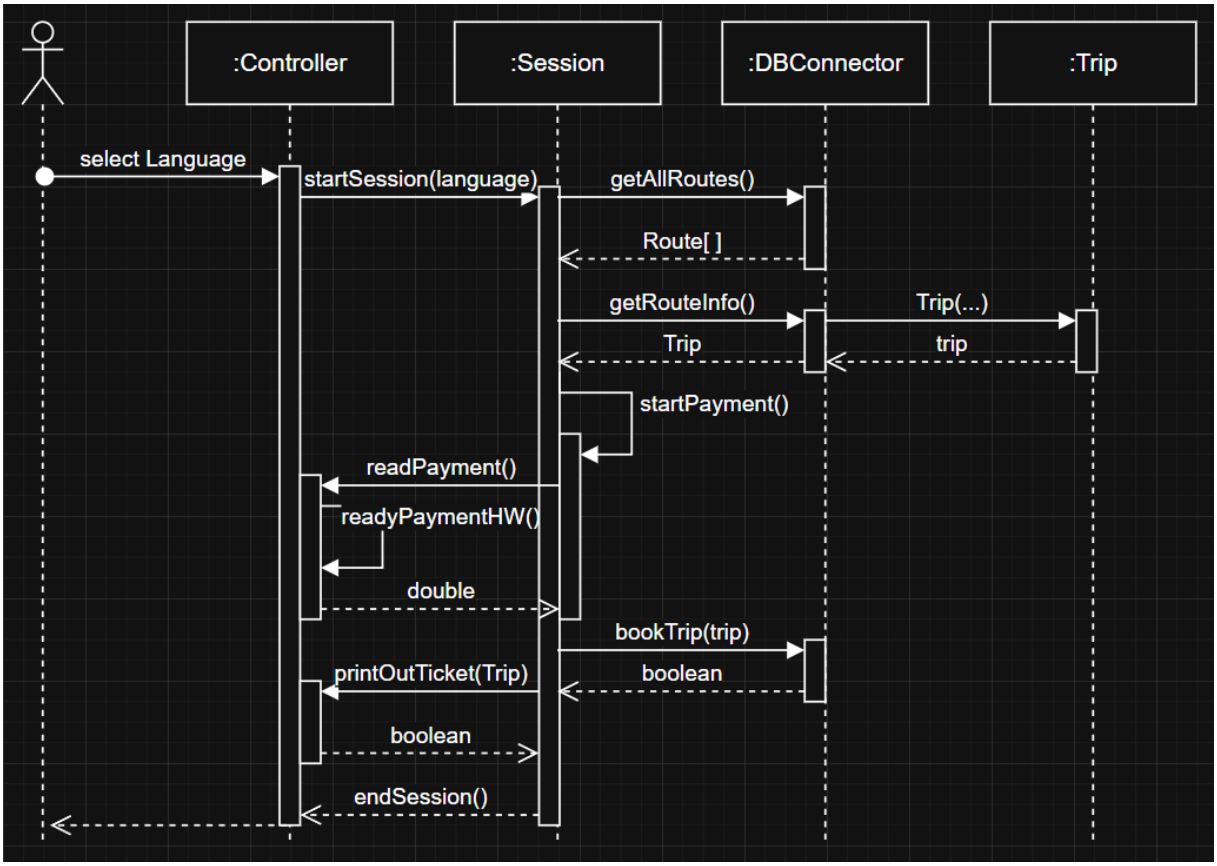
Logical View:



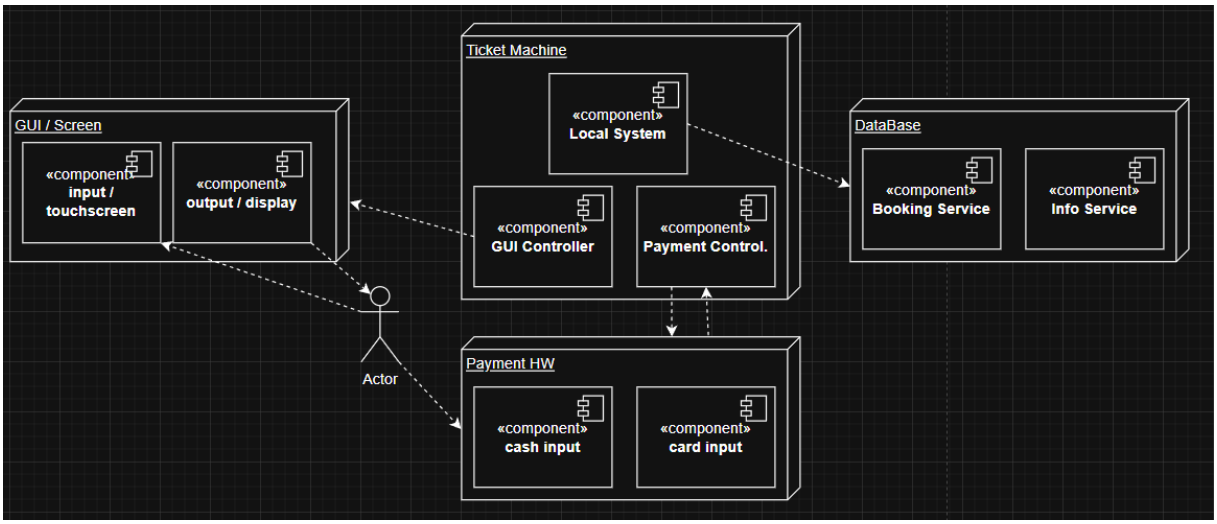
Development View:



Proccess View:



Physical View:



4.

a) A whistleblowing system on the internet.

   Data-Centric / Event-based -> acts as an anonymous *Blackboard / Message Queue*

b) A video conferencing system

   Hierachical / Data-Centric / Service-oriented -> Each client sends their own stream, chat, files onto 1 Main server, which multicasts it to the other users.

c) A GPS tracker for cats

   Dataflow-centric / DistSys / Hierachical-> each collar sends a GPS Signal to the registered App / Device